ISIS Neutron and Muon Source

Mantid in a Browser: a greener alternative to cloud servers?

Duc Le

ISIS

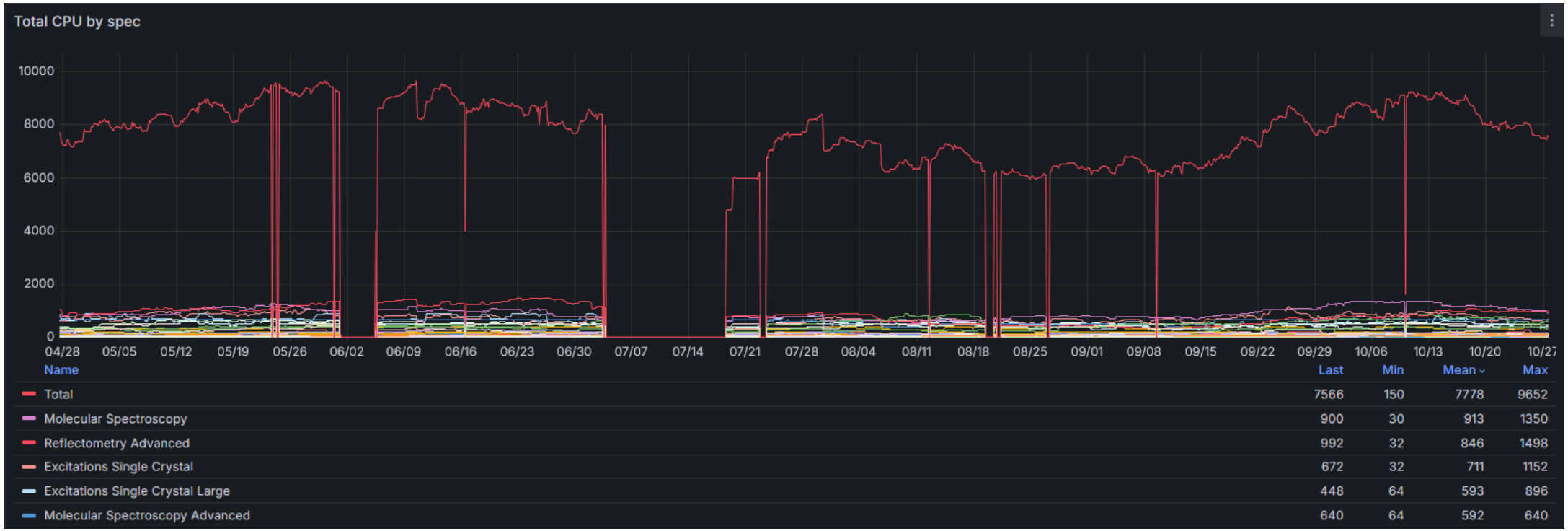
04/11/2025

(Slido)

Try it!

Introduction and Motivation

- STFC provides a cloud-based data analysis cluster accessible in a webpage via noVNC.
- ISIS used on average ~8000 cores (of ~10000) in last cycle.



Introduction and Motivation

- STFC pro
- ISIS used

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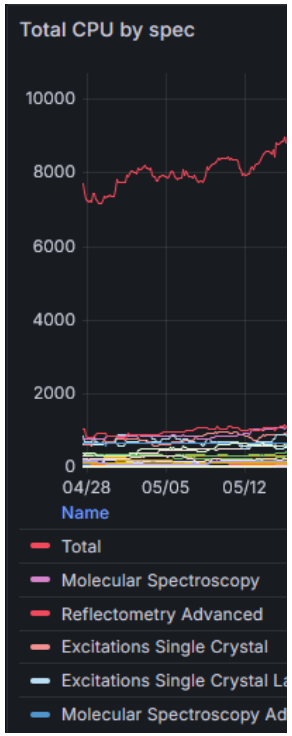

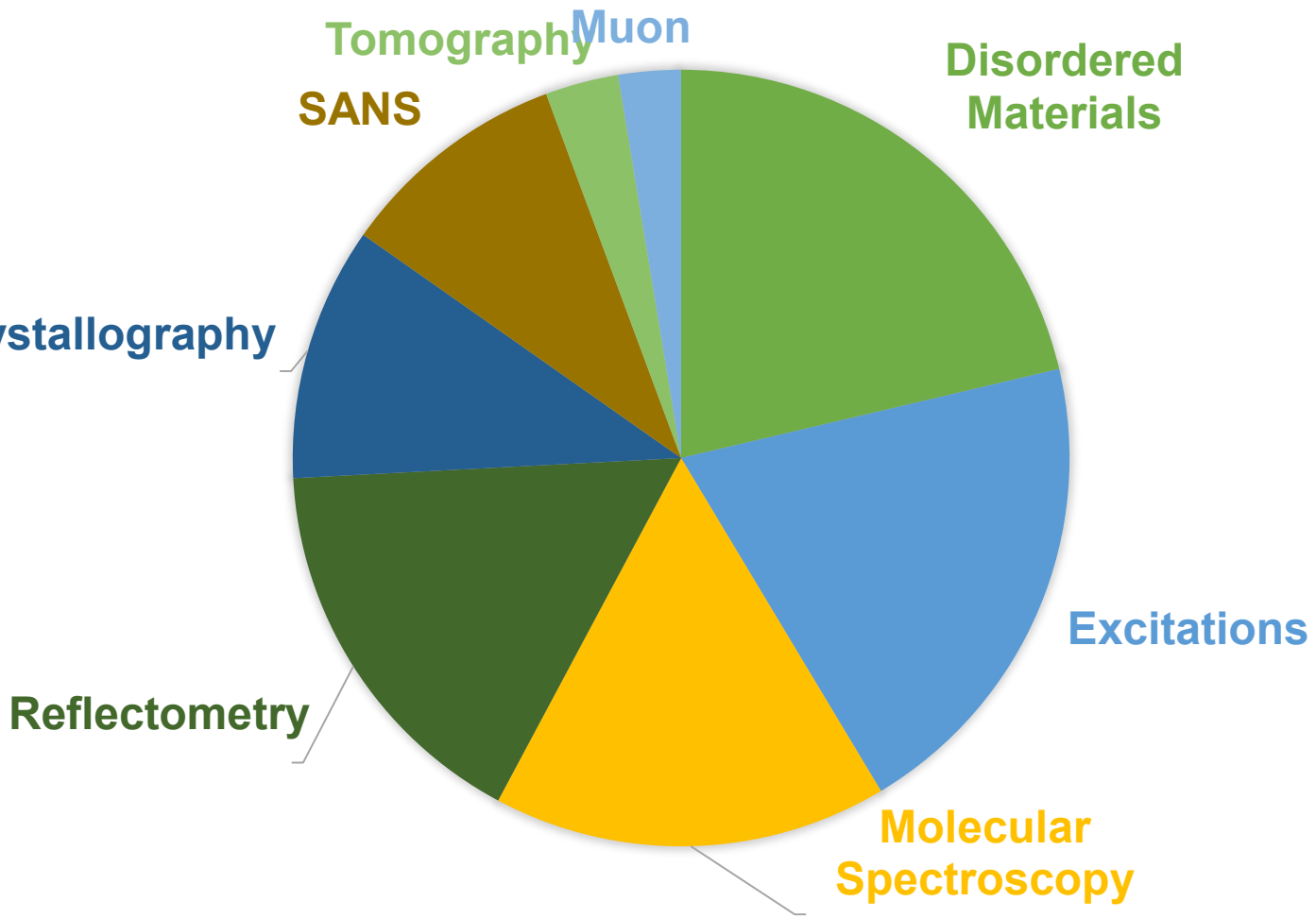

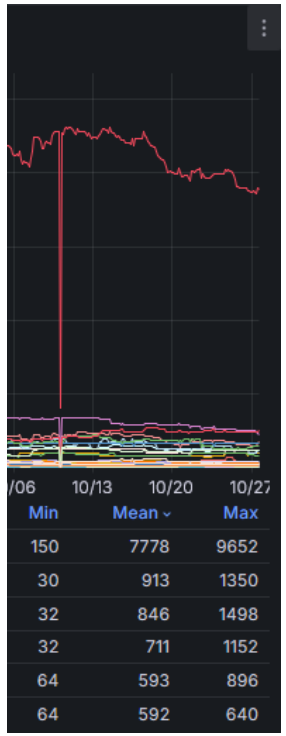

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Science and Technology Facilities Council

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- Generally users create a virtual machine (VM) for one task and then do not delete it, or have several VMs running (in separate browser tabs) each for single specific task.
- Many tasks do not require a lot of CPU/RAM but the VMs are specified for the most compute intensive tasks which are only needed some of the time.

- What if users can run some of the less intensive tasks (e.g. experiment planning) in a browser tab just like the cloud system but with the computation done **on their own computer**?

PYODIDE

WA

WebAssembly

- Modern browsers have **two** built-in interpreters: for Javascript *and WebAssembly*.
  - NB: Javascript is a 32-bit platform so is limited to 4GB memory.
- C/C++/Fortran/Rust etc. code can be compiled to WebAssembly to run in a browser.
- The cross-compiling toolchain is *Emscripten*, based on LLVM.

- Pyodide is a WebAssembly compiled version of CPython, allowing Python to be run in the browser.

- It has an ecosystem of packages (wheels) and provides a build environment on top of Emscripten but only for PEP518-compliant projects (needs `pyproject.toml`).
- Pyodide/wasm wheels are currently not accepted on PyPI – but this is working its way through PEPs and likely will be approved in future.
- Each specific version of Pyodide must use a specific version of Python and Emscripten.
  - We use Pyodide-0.27.* which targets Python 3.12 and Emscripten 3.1.58
  - Newer Pyodide-0.28.* targets Python 3.13 and Emscripten 4.0.9

WebAssembly

- Emscripten also provides a runtime environment which mimics a POSIX system.
  - There is a virtual file system, and implementations of `libc`, `libcxx` etc.
  - But it has only a rudimentary dynamic linking system: Each program must consist of a single “main module” (Pyodide/CPython for us - which is automatically linked to `libc`) and dynamically loadable “side-module”s.
  - On loading, all the symbols of the side-modules are connected to the main module – so if two side modules define the same symbol, a `duplicate symbol` runtime error is raised.
  - Side modules also increases the overheads such that often Emscripten apps are compiled statically.
  - Exceptions handling currently requires a JS module (slow). A new wasm exceptions engine is available but only in Pyodide 0.28.*
  - There is pthread support for parallelism, but it was too much of a pain to compile TBB so we’ve disabled thread parallelism in micromantid.

Micromantid

- Cannot use the Mantid build system as it is – need to compile it from pyproject.toml
- Use `scikit-build` instead and almost identical CMakeFiles:
  - We choose to build a single shared-object: `_micromantid.so` instead of the set of `_kernel.so`, `_api.so`, `_dataobject.so`, `_geometry.so`, and `_curvefitting.so`.
  - But wasm has a hard max limit of 10000 exported symbols which we are close to hitting.
- Support libraries (`boost`, `poco`, `gsl`, `libhdf5`) are compiled as separate share-object libraries ("side-modules") and bundled into the wheel. (TODO: TBB)
- Mantid factory class registrations uses a `#define` macro with a dummy class whose constructor has the side-effect (as the result of the "comma operator") of calling the register method of the factory.
  - This doesn't work in Emscripten (don't know why not) – we need to call register on all algorithms, dataobjects, fit functions etc. manually… ☹
- Poco::Net doesn't work in Emscripten (need to call web APIs directly… subject to a bunch of restrictions) – so have to disable archive, config, help and usage services.
- Also need small changes for Python 3.12 (just in `funcinspect.py`) and Numpy 2.

Micromantid

- Project has main mantid repo as a git submodule – changes to the base code are kept as a set of `diffs`.
- We also have to bundle instrument files and default `Mantid.properties` in the wheel (also means that changes to these are hard for users to make).
- Have a wrapper over the Python tests in `Framework/PythonInterface/test/python/mantid` which is run in `node.js` in github-actions – `micromantid` passes 125 out of 131 tests. CI build/test job takes ~1h.

Inelastic Neutron Scatering Web Analysis Environment (Inswae)

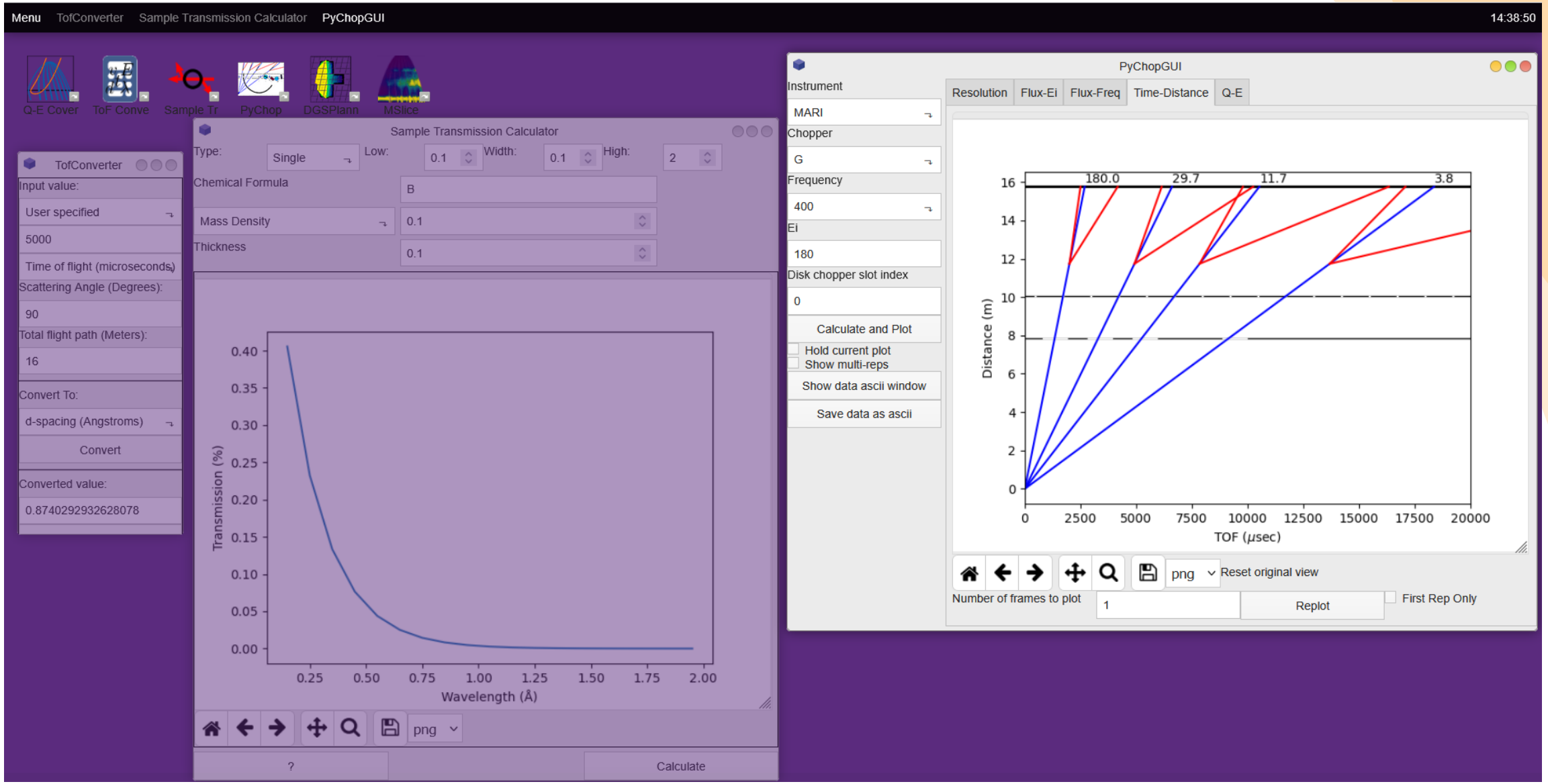
- Mantid GUI uses Qt in C++ (Workbench, several interfaces), or PyQt5.
- It is possible to compile Qt for wasm but is quite difficult, and PySide/PyQt is not fully supported.
- Mantid only uses GUI components and not the bulk of Qt's functionalities

- Solution: write a Python-Javascript shim to handle the graphical components only (e.g. we don't need the webkit browser, etc.)

- Use `os.js` as the windowing manager (it is based on the `hyperapp` lightweight framework).
- Overrides `qtpy` namespace, mostly wrapping standard `html` widgets.

- Allows to port Python interfaces to run in the browser:
  - `ToFConverter`,
  - `QECoverage`,
  - `PyChop`,
  - `SampleTransmissionCalculator`,
  - `DGSPlanner`,
  - `MSlice`

- As such, cannot run C++ interfaces at the moment (need a shim for C++ Qt to html/js).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Total flight path (Meters):
16
Convert To:
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Convert
Converted value:
0.8740292932628078
Sample Transmission Calculator
Type:
Single
Low:
0.1
Width:
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2
Chemical Formula
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PyChopGUI
Instrument
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400
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180
Disk chopper slot index
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Show data ascii window
Save data as ascii
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Time-Distance
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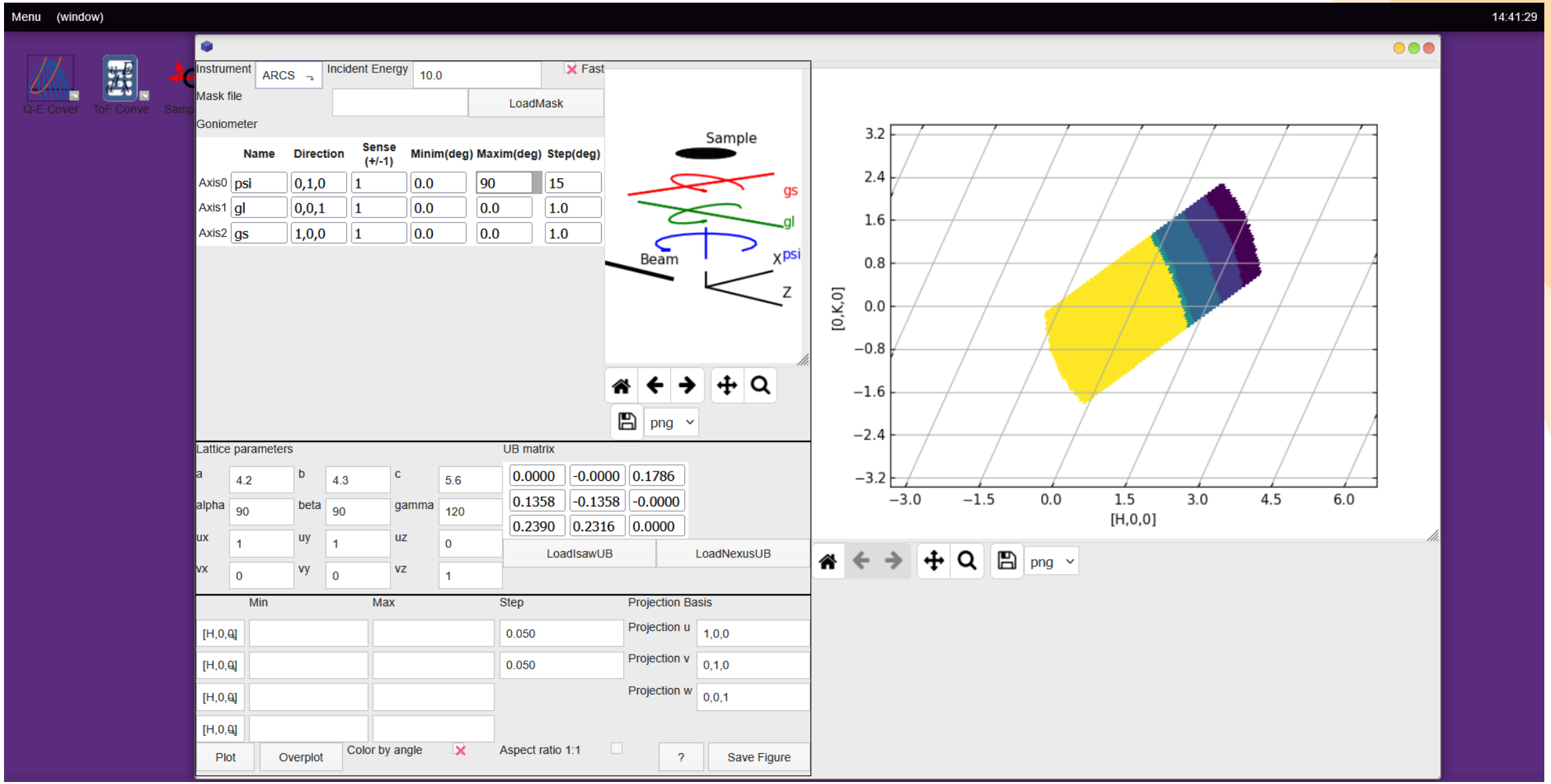
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Axis2 gs 1,0,0 1 0.0 0.0 1.0
Sample
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png
Lattice parameters
a 4.2
b 4.3
c 5.6
alpha 90
beta 90
gamma 120
ux 1
uy 1
uz 0
vx 0
vy 0
vz 1
UB matrix
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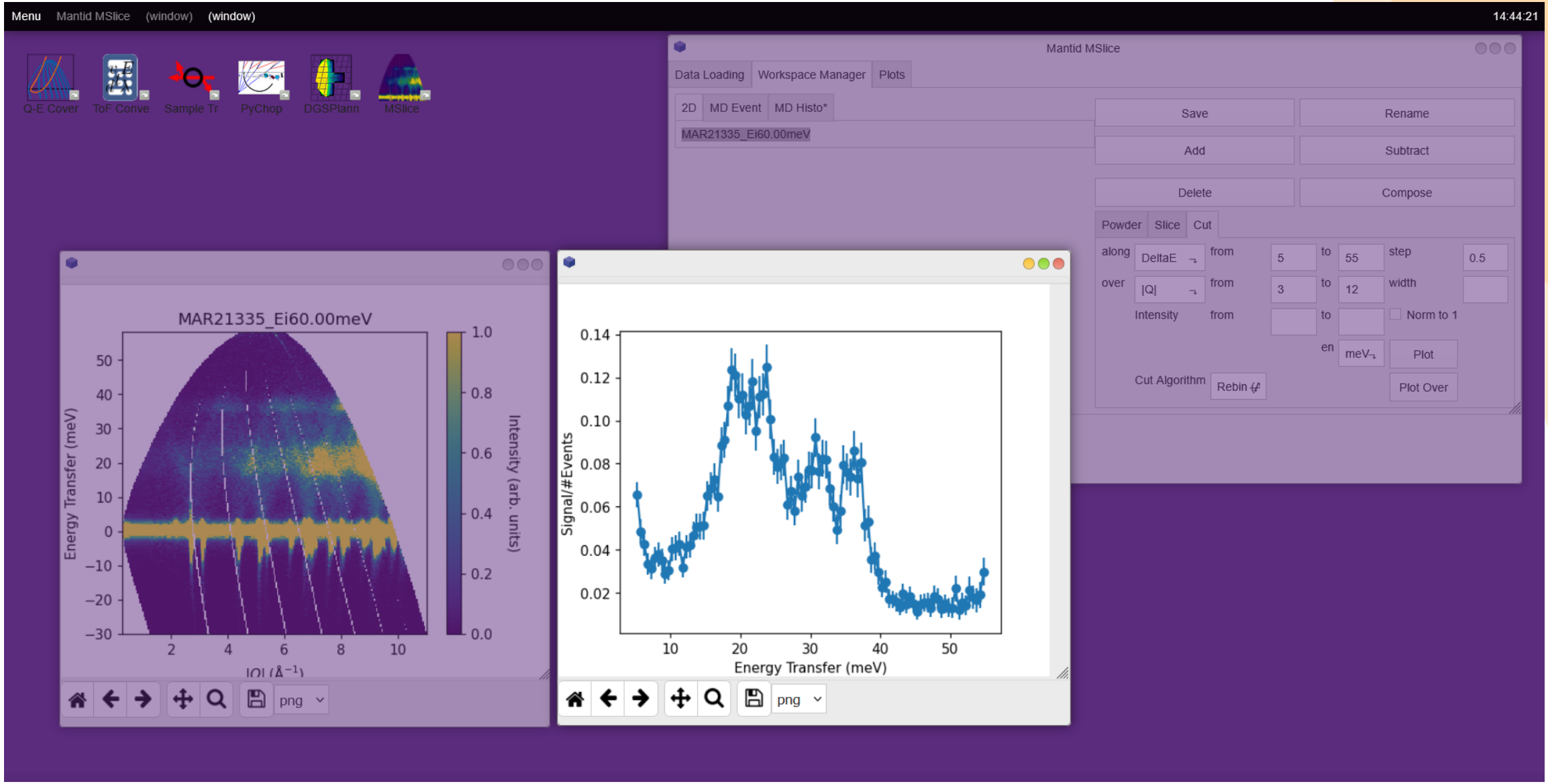
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Subtract
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Compose
Powder
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width
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from
to
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en
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Plot
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png
Signal/#Events
Energy Transfer (meV)
png

Future

- Add TBB library and re-enable threading in micromantid
- Make a jupyterlite version
- C++/wasm Qt wrapper

Questions?

Try it!